

Visual Output

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Reference: M. Margolis, et al., Arduino Cookbook, 3rd Edition,
Oreilly, 2020

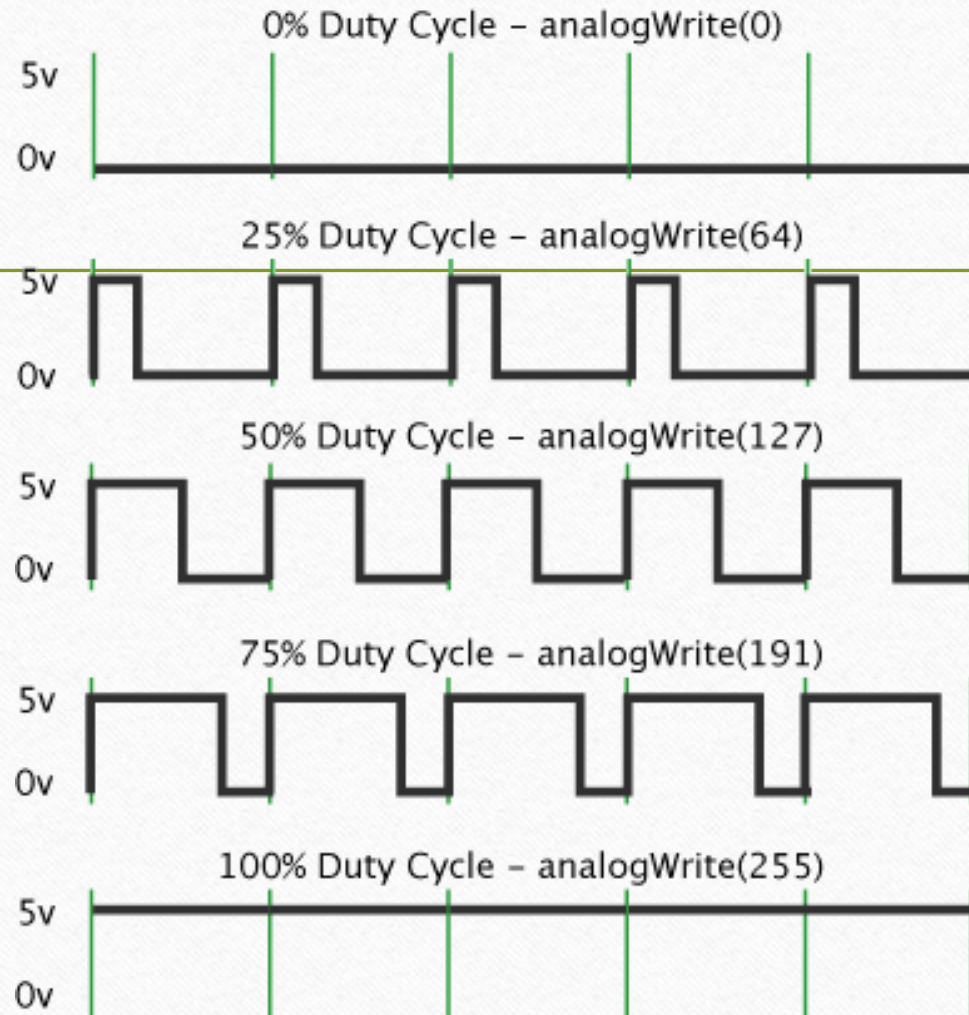
Digital Output

- All the pins that can be used for digital input can also be used for digital output.
- `digitalWrite(outputPin, value)`
 - `outputPin`: the pin to control
 - `value`: HIGH (5 or 3.3 volts) or LOW (0 volts)
 - The pin must have been set in *output* mode using the *pinMode* command.

Analog Output

- analogWrite
 - not truly analog
 - uses a technique called **Pulse Width Modulation (PWM)** to emulate an analog signal using digital pulses.
- PWM
 - varies the proportion of the pulses' on time and off time
 - Low-level output: produces pulses that are **ON** more short period of time
 - High-level output: produces pulses that are **ON** more than they are off.

Pulse Width Modulation



LED not lit

LED $\frac{1}{4}$ brightness

LED $\frac{1}{2}$ brightness

LED $\frac{3}{4}$ brightness

LED full brightness

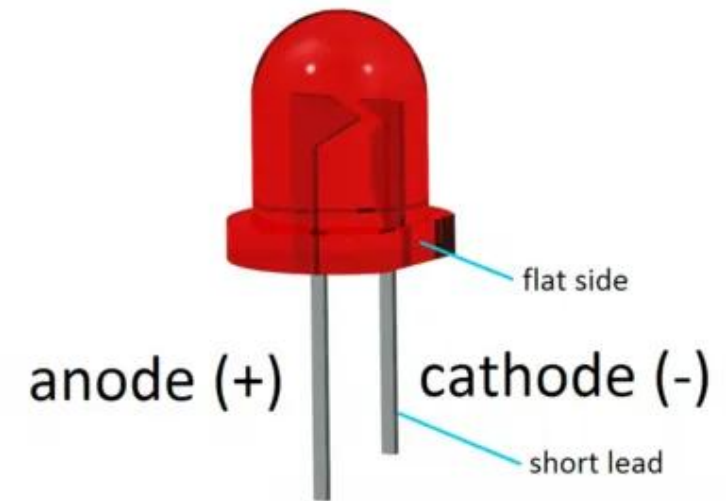
PWM pins

- Arduino has a limited number of pins that can be used for PWM output.
- Arduino Uno
 - pins 3, 5, 6, 9, 10 and 11.



Controlling Light

- An LED is a diode (二極體) with two leads (導線)
 - an anode (陽極)
 - usually the longer lead
 - a cathode (陰極)
 - often a flat spot on the housing
- When the voltage on the anode is more than that on the cathode (by an amount called the *forward voltage*), the device emits light (photons).
 - A typical red LED has a forward voltage of around 1.8 volts.



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- When the voltage on the anode becomes 1.8 volts more positive than that on the cathode, the LED “turns on” and effectively becomes **a short circuit**.
 - Must limit the current with a resistor.
 - or the LED will (sooner or later) burn out.
 - Consult an LED datasheet to determine values for forward voltage and maximum current.

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- Arduino pins on Uno boards can supply up to 40 mA of current.
 - not enough to drive the higher-brightness LEDs or multiple LEDs connected to a single pin.
 - May use a transistor to increase the current through the LED.
 - Multicolor LEDs consists of two or more LEDs in one physical package.

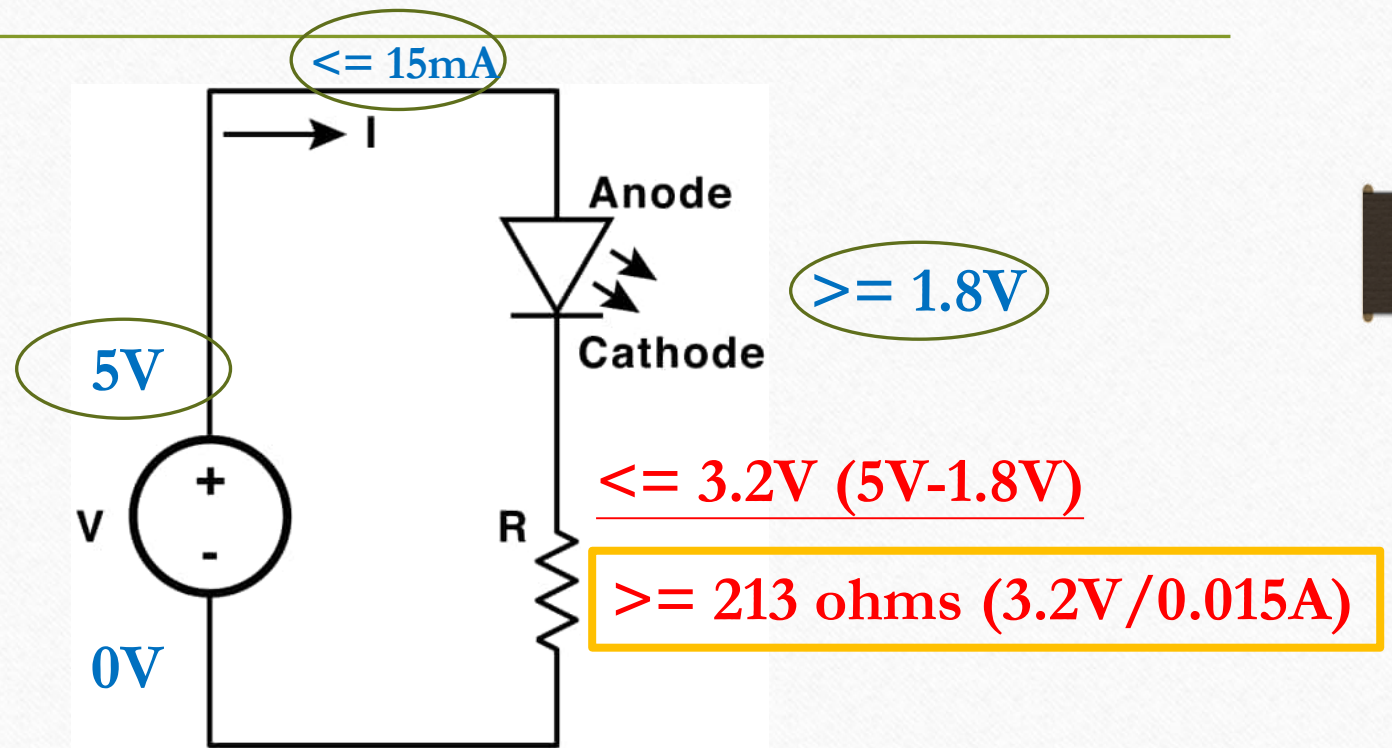
Maximum Pin Current

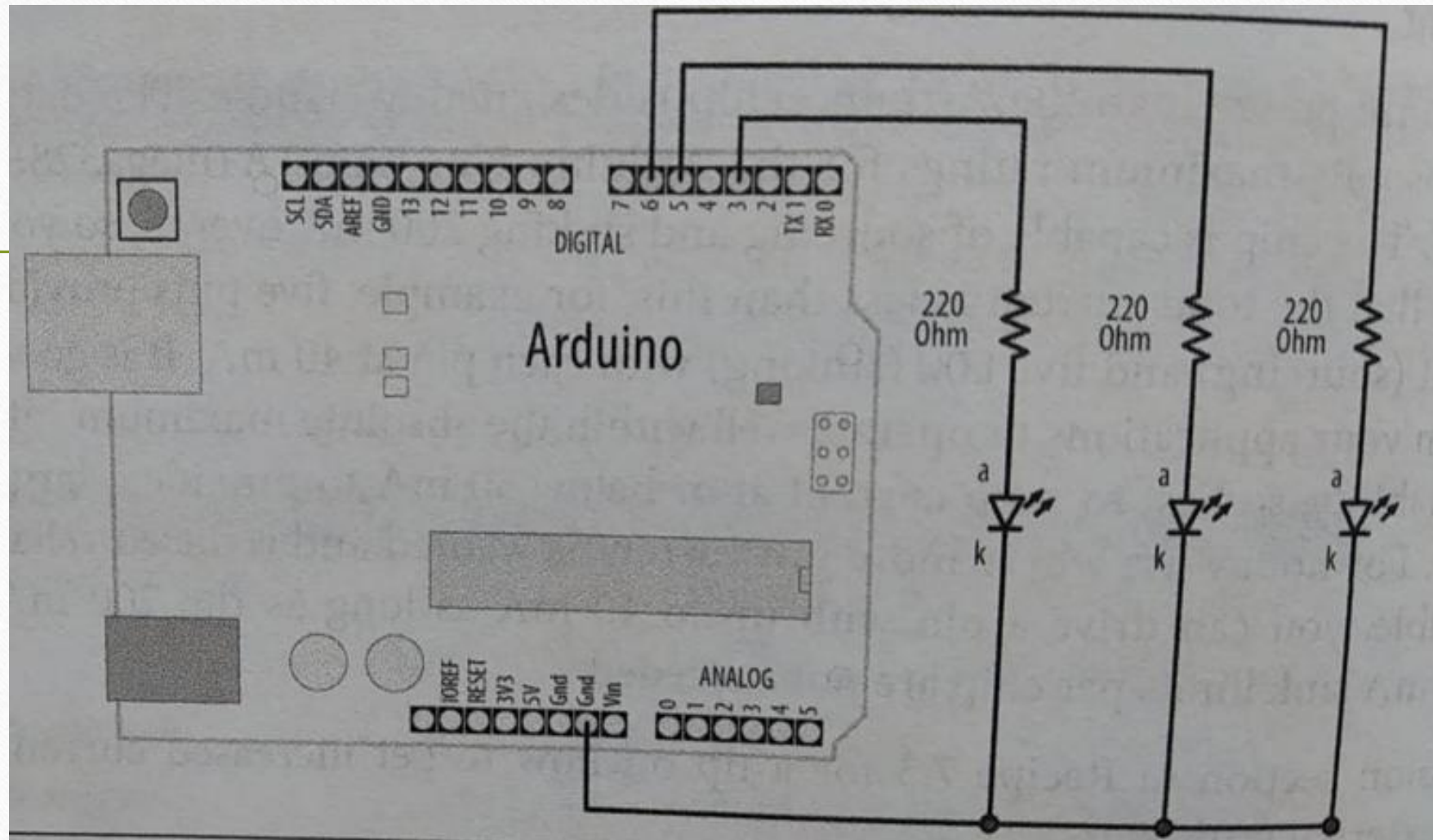
- LEDs can draw more power than the Arduino chip is designed to handle.
- Arduino Uno
 - the absolute maximum ratings for the Arduino Uno chip as 40 mA per pin.
 - The chip is capable of sourcing and sinking 200 mA overall.
 - You must ensure that the total current is less than this
 - For instance:
 - five pins providing a HIGH output (sourcing) and five LOW (sinking) with each pin at 40 mA.

Calculate the Resistor Value to Control the Current through an LED

- Need to know
 - the input power supply voltage (V_S)
 - the LED forward voltage (V_F)
 - the amount of current (I) that you want to flow through the LED
- Ohm's law
 - $V = IR$
 - $R = V/I = (V_S - V_F) / I$

- $V_S = 5V$
- $V_F = 1.8V$
- $I = 0.015A$ (15mA)





Adjusting the Brightness of an LED

- Connect each LED to an analog (PWM) output.
- Fade the LED(s) from off to maximum intensity and back to off.

```
const int firstLed    = 3; // specify the pin for each of the LEDs
const int secondLed   = 5;
const int thirdLed    = 6;
```

```
int brightness = 0;
```

```
int increment    = 1;
```

```
#define MAX_BRIGHTNESS 255
```

```
void setup()
```

```
{
```

```
    // pins driven by analogWrite do not need to be declared as outputs
```

```
}
```



```
void loop()
{
  if(brightness > MAX_BRIGHTNESS)
  {
    increment = -1; // count down after reaching MAX_BRIGHTNESS
  }
  else if(brightness < 1)
  {
    increment = 1; // count up after dropping back down to 0
  }
  brightness = brightness + increment; // increment (or decrement sign is minus)

  // write the brightness value to the LEDs
  analogWrite(firstLed, brightness);
  analogWrite(secondLed, brightness);
  analogWrite(thirdLed, brightness );

  delay (10); // 10ms for each step change means 2.55 secs to fade up or down
}
```